NVIDIA GeForce RTX 4090 Technical Brief - DATASHEET

Manufacturer: NVIDIA
Part Number: RTX-4090
Category: GPU / Consumer GPU
Document Revision: Rev 1.0
Language: en

Overview

The NVIDIA GeForce RTX 4090 is a flagship GPU based on the Ada Lovelace architecture (AD102). It provides 16,384 CUDA cores, 4th-generation Tensor Cores with FP8 support and 3rd-generation RT cores. 24 GB GDDR6X memory. Suited for workstation AI development, rendering and content creation; not designed for data center multi-GPU deployments.

Thermal and Power

TDP: 450W. Recommended system power: 850W. Power connector: 1x 16-pin 12VHPWR. Boost clock: 2520 MHz. Memory interface: 384-bit, 1008 GB/s bandwidth. PCIe Gen4 x16. Maximum GPU temperature: 90 degrees C. Triple-slot cooler. No NVLink support. Outputs: HDMI 2.1a, 3x DisplayPort 1.4a.

Key Specifications Summary

memory: 24GB_GDDR6X. bw: 1008GBs. cuda: 16384. tensor: 512. rt: 128. tdp: 450W. pcie: gen4.